Prompt Engineering & AI Response Quality Comparison Study

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Project Type: Independent Portfolio Project
Number of Tasks: 30
Total AI Responses: 60
Evaluation Scale: 1–5
AI Model Evaluated: Chat GPT

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1. Executive Summary

This project investigates how prompt design influences the quality of AI-generated responses.

A total of **30 tasks across 10 categories** were selected. Each task was evaluated using two versions of a prompt:

1. An initial prompt containing relatively basic instructions.
2. An improved prompt containing additional task-specific instructions, context, formatting requirements, audience specifications, examples, or output constraints.

Both prompt versions were submitted to the same AI model, producing **60 AI-generated responses**.

The responses were evaluated using a standardized rubric covering:

- Relevance
- Completeness
- Clarity
- Instruction Following
- Specificity
- Overall Quality

The project then compared the initial and improved responses to determine whether structured prompt improvements resulted in measurable improvements in response quality.

> The average overall quality score increased from **[3.5/5]** for initial prompts to **[4.2/5]** for improved prompts, representing an average improvement of **[0.7] points**.

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2. Project Objective

The primary objective of this project is to investigate how prompt engineering techniques affect the quality and usefulness of AI-generated responses.

The project aims to:

1. Design 30 real-world AI tasks.
2. Create an initial prompt for each task.
3. Generate AI responses using the initial prompts.
4. Identify weaknesses or limitations in the initial prompts.
5. Create improved versions of the prompts.
6. Generate new AI responses using the improved prompts.
7. Compare the initial and improved responses.
8. Evaluate the responses using a standardized rubric.
9. Identify which prompt-engineering techniques produced improvements.
10. Document cases where prompt improvements were unsuccessful or produced limited benefits.

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3. Research Question

The main research question is:

> **How does adding explicit instructions, context, formatting requirements, audience specifications, and output constraints affect the quality and instruction-following performance of AI-generated responses?**

Additional questions include:

- Do improved prompts produce higher-quality responses?
- Which prompt-engineering techniques produce the greatest improvements?
- Does specifying the intended audience improve clarity?
- Does specifying an output format improve instruction following?
- Do word-count requirements improve response quality?
- Can excessive instructions negatively affect the response?
- Are there situations where the original prompt performs equally well or better?

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4. Dataset

The dataset consists of **30 AI-related tasks** covering 10 categories.

Categories

Category	Number of Tasks
General Knowledge	3
Science	3
History	3
Technology	3
Business	3
Education	3
Writing	3
Reasoning	3
Instruction following	3
Ambiguous Questions	3
Total	**30**

Each task contains:

- Task ID
- Category
- Task
- Initial prompt
- Initial AI response
- Initial scores
- Improved prompt
- Improved AI response
- Improved scores
- Evaluation scores
- Prompt-engineering technique used

5. Experimental Design

Each task was tested using two prompt versions.

Stage 1 — Initial Prompt

The initial prompt was designed to represent a relatively simple or less-constrained user request.

Example:

> Explain gravity.

Stage 2 — Improved Prompt

The initial prompt was redesigned using one or more prompt-engineering techniques.

Example:

> Explain gravity to a 12-year-old using simple English, one everyday example, and no more than 150 words.

Stage 3 — Response Generation

Both prompts were submitted to the same AI model.

This produced:

30 initial responses + 30 improved responses = 60 responses

Stage 4 — Evaluation

Both responses were evaluated using the same evaluation rubric.

Stage 5 — Comparison

The scores and qualitative observations were compared to identify whether the improved prompt produced a better response.

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6. Prompt Engineering Techniques

The improved prompts used different techniques depending on the task.

6.1 Audience Specification

The prompt explicitly identifies who the response is intended for.

Example:

> Explain machine learning to a beginner.

This can help the model adjust vocabulary and explanation complexity.

6.2 Output Format Specification

The prompt specifies how the answer should be structured.

Example:

> Compare Python, SQL, and Excel in a table.

This allows instruction-following performance to be evaluated more clearly.

6.3 Length Constraints

The prompt specifies the desired response length.

Example:

> Explain photosynthesis in approximately 100 words.

This tests whether the AI can follow explicit length requirements.

6.4 Explicit Requirements

The prompt specifies information that must be included.

Example:

> Include sunlight, water, carbon dioxide, glucose, and oxygen.

This reduces ambiguity about the expected content.

6.5 Context

Additional information is provided to clarify the purpose of the task.

Example:

> Explain cloud computing to someone with no technical background.

6.6 Structured Instructions

Multiple requirements are organized into a clear instruction.

Example:

> Define the concept, explain how it works, provide two examples, and summarize the main benefits.

6.7 Examples

Where appropriate, examples can be included to demonstrate the desired type of output.

This technique can help guide the model toward a particular response pattern.

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7. Evaluation Methodology

Each response was evaluated using a standardized 1–5 scoring rubric.

The evaluation criteria were:

Criterion	Maximum Score
Relevance	5
Completeness	5
Clarity	5
Instruction Following	5

Specificity	5
Overall Quality	5
Total	**30**

The same criteria were applied to both the initial and improved responses.

This ensures that the comparison is based on consistent evaluation standards.

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8. Evaluation Criteria

Relevance

Measures whether the response directly addresses the requested task.

Completeness

Measures whether the response covers the important requirements of the prompt.

Clarity

Measures how understandable, organized, and readable the response is.

Instruction Following

Measures whether the AI followed explicit requirements such as format, length, audience, tone, and requested content.

Specificity

Measures whether the response provides sufficiently precise and useful information rather than vague statements.

Overall Quality

Represents the evaluator's overall assessment of the usefulness and quality of the response.

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9. Scoring Method

Each response receives a score from 1–5 for each criterion.

Therefore:

Maximum score per response = 30

The total score can be calculated as:

> **Total Score = Relevance + Completeness + Clarity + Instruction Following + Specificity + Overall Quality**

For example:

Criterion	Score
Relevance	5
Completeness	4
Clarity	5
Instruction Following	4
Specificity	4
Overall Quality	5
Total	**27/30**

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10. Results

10.1 Overall Comparison

FILL THIS SECTION AFTER EVALUATING ALL 60 RESPONSES.

Evaluation Criterion	Initial Prompts	Improved Prompts	Difference

Relevance	4	4.5	.0.5
Completeness	3.5	4.2	0.7
Clarity	3.5	4.5	1
Instruction Following	3.8	4.4	0.6
Specificity	3.9	4.3	0.4
Overall Quality	3.8	4.4	0.6

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11. Improvement Analysis

For each task, calculate:

> **Improvement = Improved Overall Score − Initial Overall Score**

For example:

Task	Initial	Improved	Improvement
001	3	5	+2
002	4	5	+1
003	3	4	+1

These numbers are only examples.

Overall Improvement

FILL AFTER COMPLETING

> Across the 30 tasks, the average overall score changed from **[3.8]** to **[4.4]**, resulting in an average improvement of **[0.6 points]**.

12. Prompt Technique Analysis

FILL AFTER COMPLETING

Analyze which techniques produced the greatest improvements.

Prompt Technique	Tasks Using Technique	Average Improvement
specify the Audience	9	4
specify Output format	1	3.9
specify length	4	3.8
Give a role/purpose	5	4
Give Explicit requirements	4	4.5
provide Examples	4	4
add Context	2	4
add Constraint	5	4.2
defined the rule	7	4
request step-by-step explanation	1	4
handle ambiguity	1	4.5

Then explain your findings.

Example structure:

> The analysis showed that Give Explicit requirements. produced the largest average improvement. This appeared to be particularly useful for tasks requiring Explicit requirements.

13. Instruction-Following Analysis

This section focuses specifically on whether the improved prompts helped the AI follow instructions.

FILL AFTER COMPLETING

For example:

> Initial prompts achieved an average instruction-following score of **[3.8]**, while improved prompts achieved **[4.4]**.

Then discuss the types of instructions that were most successfully followed:

- specify the Audience
- Give a role/purpose
- Give Explicit requirements
- provide Examples
- add Context
- add Constraint
- defined the rule

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14. Examples of Successful Improvements

Choose **3–5 strong examples** from your actual experiment.

Example 1 — Machine learning

Initial Prompt:

> Explain machine learning.

Improved Prompt:

> Explain machine learning to a beginner using simple English. Include a definition, how it works at a high level, and two everyday examples.

Initial Score: 3.6/5
Improved Score: 4.4/5

Analysis:

In improved prompt specify what is the level of user on this topic and ask for example and asking for simple language.

Example 2 — GPS

Initial Prompt:

> Explain how GPS works.

Improved Prompt:

> What is GPS ? explain the GPS in detail to 12 year old kid in simple word, explain what is the use of it and how GPS works.

Initial Score: 3.8/5
Improved Score: 4.5/5

Analysis:

> Specify the audience and define the language and give explicit requirement.

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15. Unsuccessful Improvements

This section is extremely valuable because **not every prompt improvement should work**.

Find tasks where:

> Improved Score ≤ Initial Score

Then investigate why.

Possible explanations include:

- Too many instructions
- Conflicting requirements
- Excessive constraints
- Unclear wording
- Unnecessary context
- The original prompt was already sufficiently clear
- The AI misunderstood a new instruction
- The added constraint reduced useful information

Example

Initial Prompt:

> How can students improve their memory?

Improved Prompt:

> Give five evidence-based strategies for improving learning and memory. Explain each strategy in 1–2 sentences and avoid making medical claims.

Initial Score: 3.8/5
Improved Score: 3/5

explanation:

> The improved prompt added multiple formatting and length restrictions. Although the response followed the requested format, the additional constraints reduced the amount of useful information provided.

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16. Key Findings

Finding 1

Improved prompts perform better overall.

Finding 2

Explicit requirements technique produced the largest improvement.

Finding 3

technology and history category benefited most.

Finding 4

Education category showed the smallest improvement.

Finding 5

In Education category the improved prompt performed worse.

Finding 6

Give Explicit requirements and Context type of instruction did the AI follow most reliably.

Finding 7

Evidence based instruction caused the most problems.

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17. Discussion

The results should be interpreted carefully.

Prompt engineering does not simply mean **adding more instructions**.

A useful prompt should provide the model with enough information to understand:

- What needs to be done
- Who the response is for
- What information is required
- What format is expected
- What constraints apply

However, adding unnecessary or conflicting instructions may not improve the result.

The purpose of this experiment is therefore not simply to determine whether longer prompts are better, but to investigate **which types of instructions are useful for specific tasks**.

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18. Limitations

This project has several limitations.

1. Limited Dataset

The experiment contains only 30 tasks, so the findings should not be treated as universally applicable.

2. Single Evaluator

The evaluation was conducted by one evaluator, which means subjective judgment may influence some scores.

3. Model Dependence

Results may differ when using another AI model.

4. Model Updates

AI models can change over time, meaning the same prompts may produce different results in the future.

5. Task Selection

The selected tasks represent only a small sample of possible AI use cases.

6. Evaluation Subjectivity

Although a standardized rubric was used, some judgments—especially clarity and overall quality—can involve subjective interpretation.

6. Evaluation category

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19. Quality Assurance

To improve consistency:

- The same evaluation criteria were used for all responses.
- Initial and improved responses were evaluated using the same scoring scale.
- Prompt versions were kept clearly separated.
- Scores were reviewed for obvious inconsistencies.
- Actual observations were separated from assumptions.
- Results were calculated from the completed dataset.

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20. Conclusion

This project examined the effect of prompt engineering on AI-generated response quality across 30 tasks. The experiment compared responses generated from initial prompts with responses generated from improved prompts containing more explicit instructions and task-specific constraints.

Based on the evaluation results, improved prompts did produce higher average response quality. The techniques that showed the strongest improvement were Give Explicit requirements, while specify Output format showed more limited benefits.

The analysis also demonstrated that effective prompt engineering is not simply about making prompts longer. The usefulness of a prompt depends on providing relevant context, clear requirements, appropriate constraints, and a well-defined expected output.

The project strengthened practical skills in prompt engineering, LLM output evaluation, data analysis, critical thinking, and AI quality assessment.

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21. Skills Demonstrated

This project demonstrates practical skills in:

- Prompt Engineering
- LLM Output Analysis
- AI Response Evaluation
- Data Collection
- Data Annotation
- Comparative Analysis
- Critical Thinking
- Research Methodology
- Experimental Design
- Data Analysis
- Technical Documentation

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22. Tools Used

Only list tools you actually used.

Example:

- ChatGPT
- Google Sheets
- Google Docs
- GitHub

PROMPT ENGINEERING & AI RESPONSE QUALITY COMPARISON STUDY

An Independent Experimental Analysis of Prompt Design and AI Response Quality

Prepared by: **ANJALI**

Project: AI prompt engineering comparison